

Crab data

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Introduction

- An unfortunate byproduct of pulp-and-paper production used to be dioxins (Credits for this example: C. J. Schwarz).
- This material was discharged into waterways with the pulp-and-paper effluent where it bioaccumulated in living organisms such as crabs.
- Newer processes have eliminated this by product, but the dioxins in the organisms takes a long time to degrade.
- For monitoring purpose, each year four crabs are captured from two monitoring stations which are situated quite a distance apart on the same inlet where the pulp mill was located.
- The liver is excised and the livers from all four crabs are composited together into a single sample.
- The dioxins levels in this composite sample is measured.
- As there are many different forms of dioxins with different toxicities, a summary measure, called the Total Equivalent Dose (TEQ) is computed from the sample.

Scientific questions:

- Is the rate of decline the same for both sites ?
- Did the sites have the same initial concentration ?

Reading the data

```
crabs<-read.csv("dioxin2.csv")
head(crabs)
```

```
##   Site Year   TEQ
## 1    a 1990 179.05
## 2    a 1991  82.39
## 3    a 1992 130.18
## 4    a 1993  97.06
## 5    a 1994  49.34
## 6    a 1995  57.05
```

```
summary(crabs)
```

```
##      Site      Year      TEQ
## Length:28      Min.   :1990  Min.   : 28.16
## Class :character 1st Qu.:1993  1st Qu.: 49.59
## Mode  :character Median :1996  Median : 70.03
##                      Mean  :1996  Mean   : 82.90
##                      3rd Qu.:2000  3rd Qu.: 99.10
##                      Max.   :2003  Max.   :188.13
```

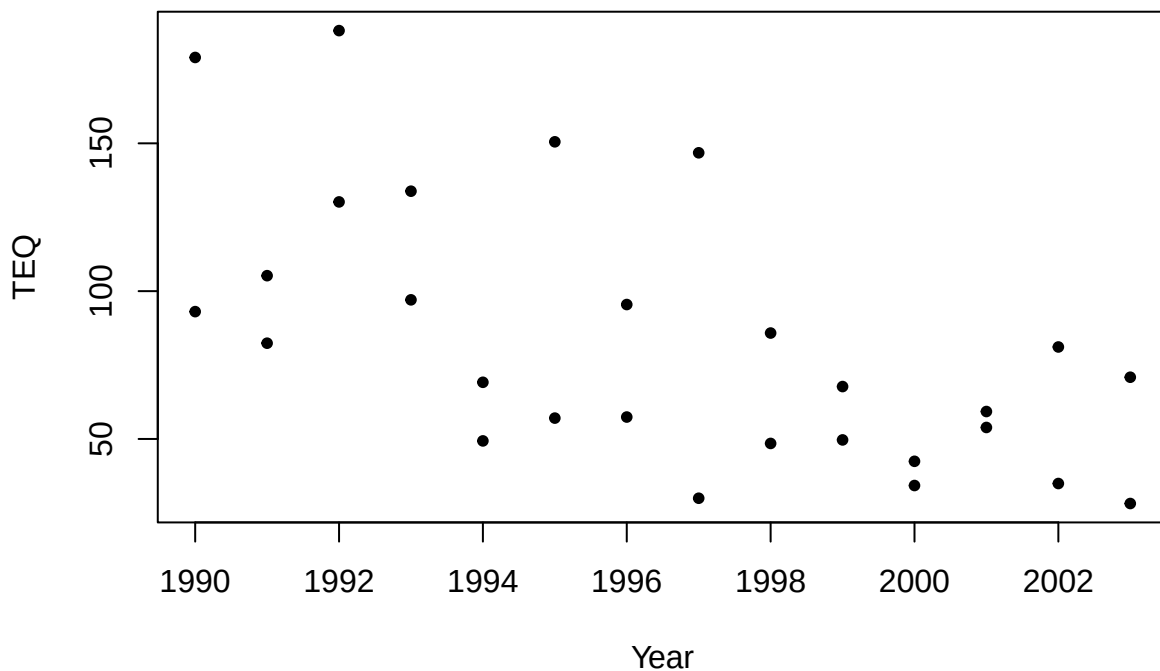
```
crabs$Site<-as.factor(crabs$Site)
```

```
summary(crabs)
```

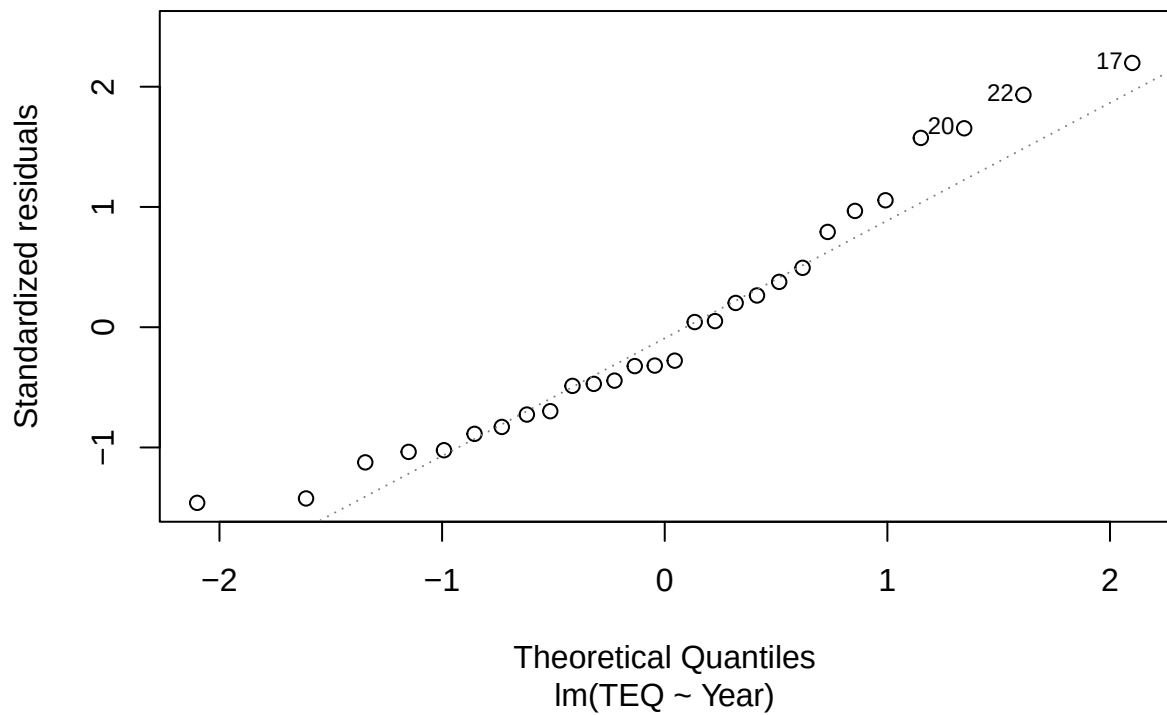
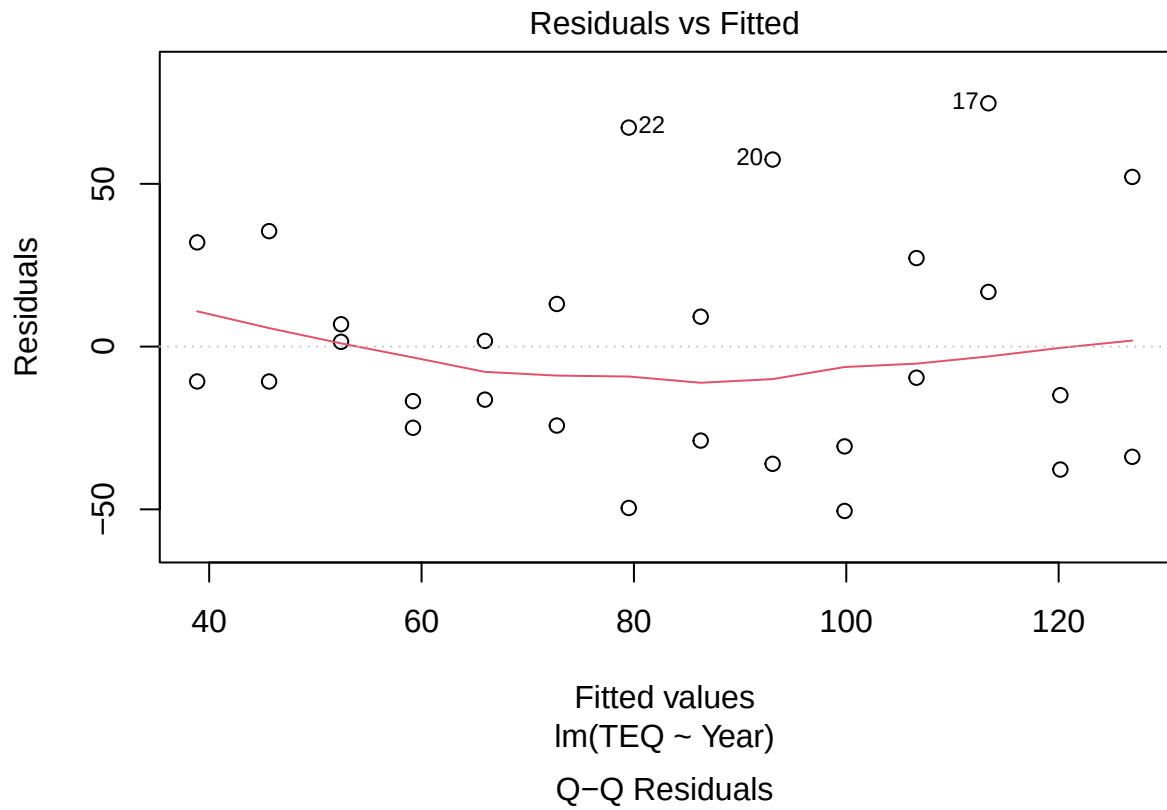
```
## Site      Year      TEQ
## a:14      Min.   :1990  Min.   : 28.16
## b:14      1st Qu.:1993  1st Qu.: 49.59
##           Median :1996  Median : 70.03
##           Mean   :1996  Mean    : 82.90
##           3rd Qu.:2000  3rd Qu.: 99.10
##           Max.   :2003  Max.    :188.13
```

Non linear relation ?

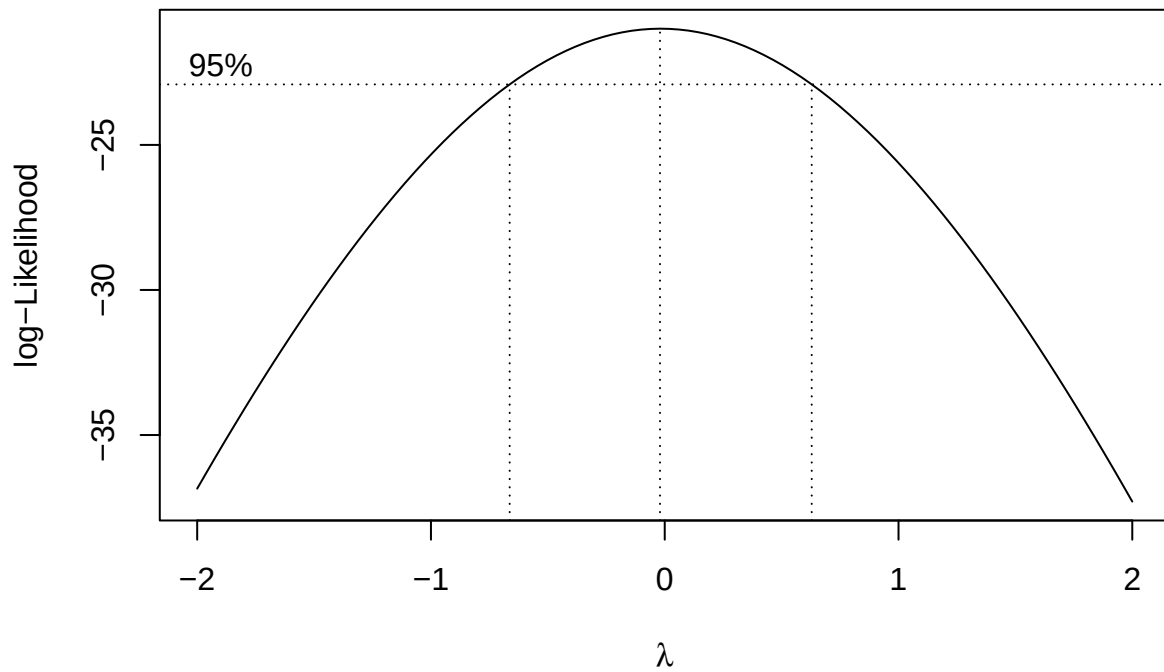
```
plot(TEQ~Year,data=crabs,pch=20)
```



```
fit.tmp<-lm(TEQ~Year,data=crabs)
plot(fit.tmp,which=c(1,2))
```

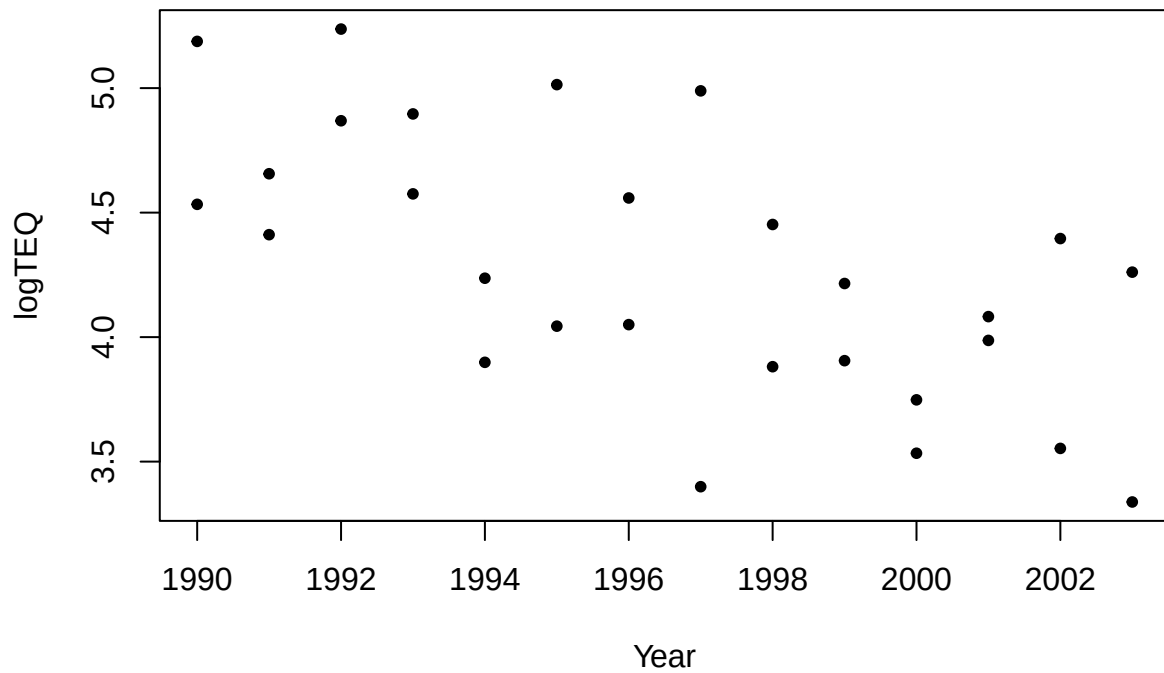


```
MASS::boxcox(TEQ~Year, data=crabs)
```

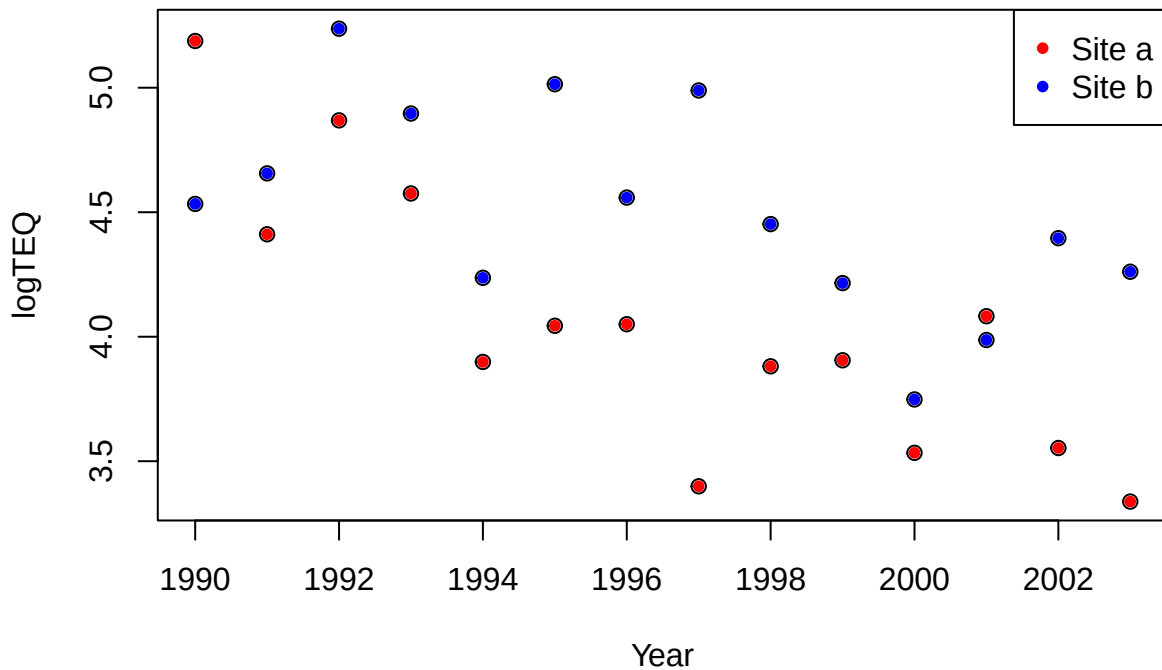


```
crabs$logTEQ<-log(crabs$TEQ)
```

```
plot(logTEQ~Year,data=crabs,pch=20)
```



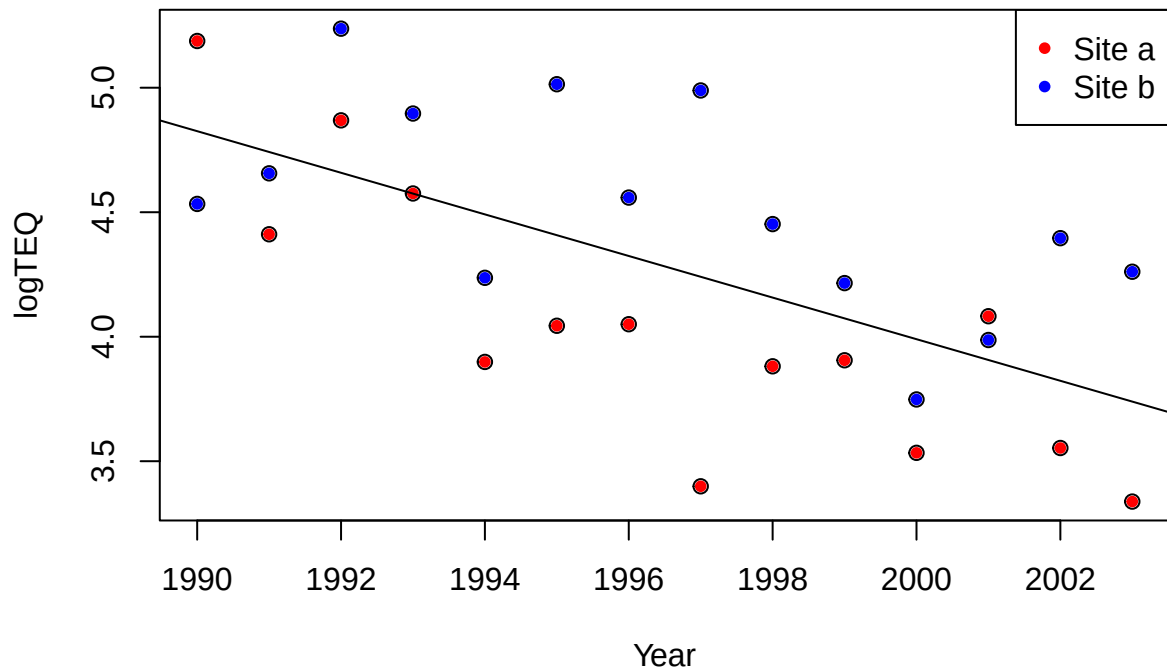
```
plot(logTEQ~Year,data=crabs)
attach(crabs)
points(Year[Site == "a"],logTEQ[Site == "a"],pch=20,col="red")
points(Year[Site == "b"],logTEQ[Site == "b"],pch=20,col="blue")
legend("topright",legend=c("Site a", "Site b"),pch=20,col=c("red", "blue"))
```



```
fit<-lm(logTEQ~Year)
summary(fit)
```

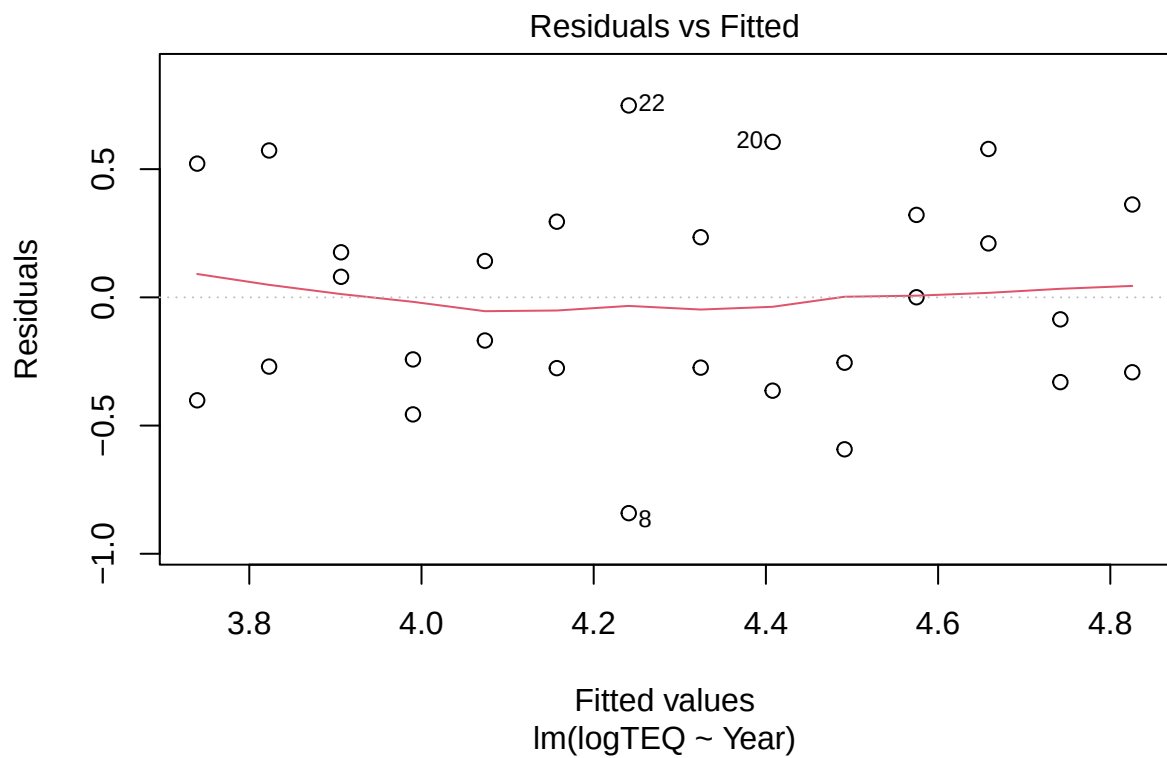
```
##
## Call:
## lm(formula = logTEQ ~ Year)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8415 -0.2801 -0.0427  0.3018  0.7483
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.07518   38.85324   4.403 0.000162 ***
## Year        -0.08354    0.01946  -4.293 0.000217 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4151 on 26 degrees of freedom
## Multiple R-squared:  0.4148, Adjusted R-squared:  0.3923
## F-statistic: 18.43 on 1 and 26 DF,  p-value: 0.0002171
```

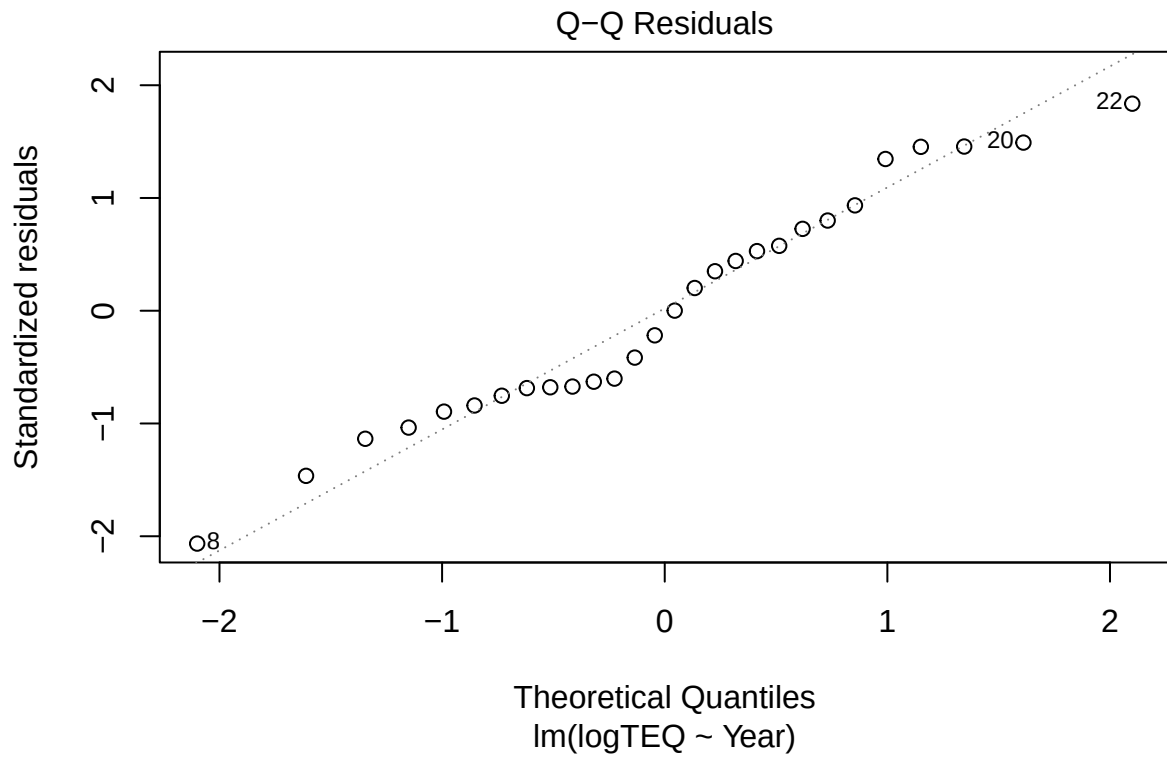
```
plot(logTEQ~Year,data=crabs)
points(Year[Site == "a"],logTEQ[Site == "a"],pch=20,col="red")
points(Year[Site == "b"],logTEQ[Site == "b"],pch=20,col="blue")
legend("topright",legend=c("Site a", "Site b"),pch=20,col=c("red", "blue"))
abline(fit)
```



Analysis of residuals

```
plot(fit,which=c(1,2))
```

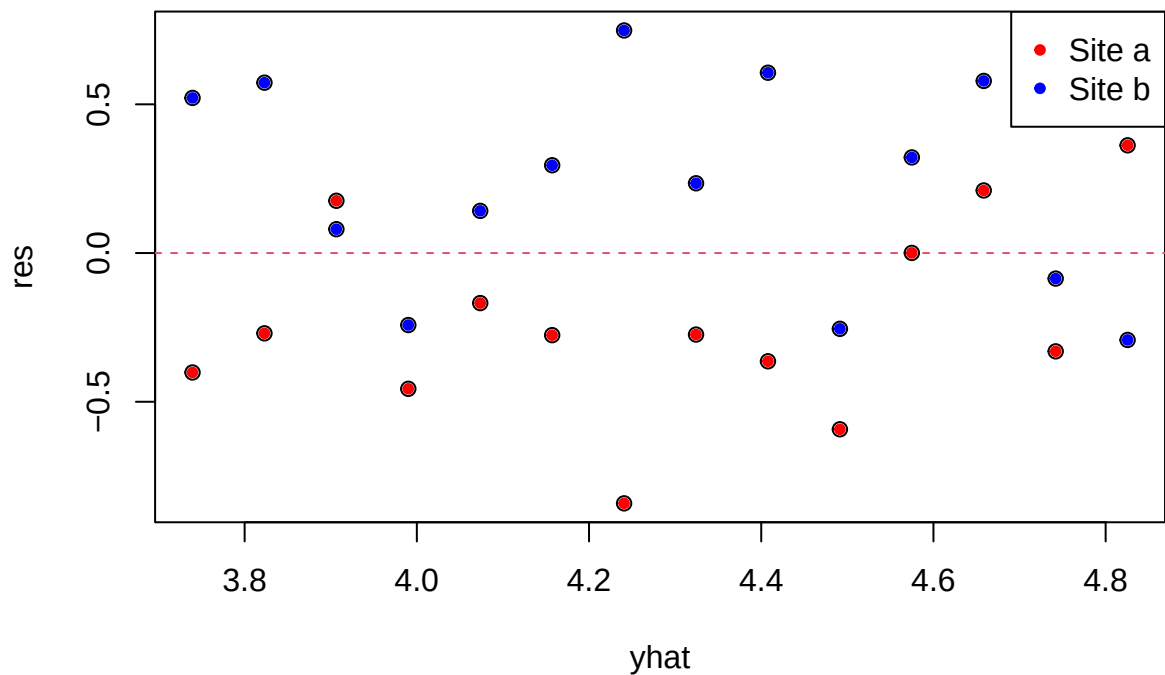




```

yhat<-fitted(fit)
res<-resid(fit)
plot(res~yhat)
abline(h=0,col=2,lty=2)
points(yhat[Site == "a"],res[Site == "a"],pch=20,col="red")
points(yhat[Site == "b"],res[Site == "b"],pch=20,col="blue")
legend("topright",legend=c("Site a", "Site b"),pch=20,col=c("red", "blue"))

```



ANCOVA models

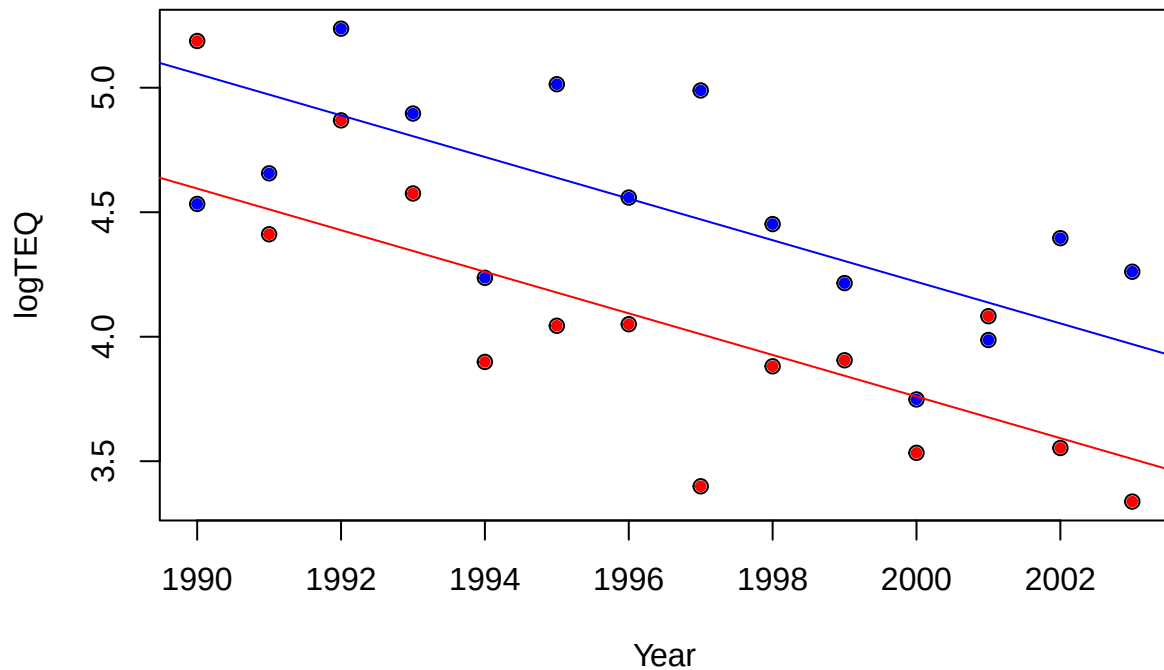
```
fit2<-lm(logTEQ~Year+Site)
summary(fit2)
```

```
##
## Call:
## lm(formula = logTEQ ~ Year + Site)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.61110 -0.18485 -0.04157  0.30391  0.59257
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 170.84475   32.38784   5.275 1.83e-05 ***
## Year        -0.08354    0.01622  -5.150 2.53e-05 ***
## Siteb        0.46086    0.13079   3.524 0.00166 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.346 on 25 degrees of freedom
## Multiple R-squared:  0.609, Adjusted R-squared:  0.5777
## F-statistic: 19.47 on 2 and 25 DF,  p-value: 7.985e-06
```

```
anova(fit,fit2)
```

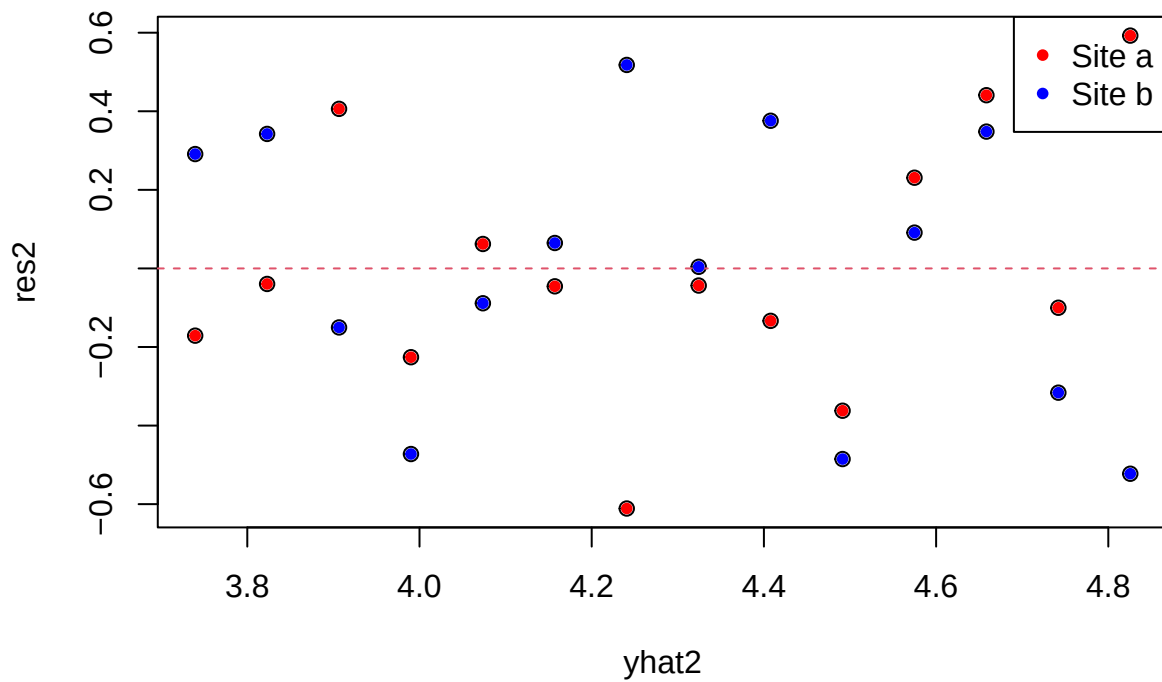
```
## Analysis of Variance Table
##
## Model 1: logTEQ ~ Year
## Model 2: logTEQ ~ Year + Site
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      26 4.4802
## 2      25 2.9935  1    1.4868 12.417 0.001663 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(logTEQ~Year,data=crabs)
points(Year[Site=="a"],logTEQ[Site=="a"],pch=20,col="red")
points(Year[Site=="b"],logTEQ[Site=="b"],pch=20,col="blue")
abline(coef(fit2)[1],coef(fit2)[2],col="red")
abline(coef(fit2)[1]+coef(fit2)[3],coef(fit2)[2],col="blue")
```



A better pattern in the residuals

```
yhat2<-fitted(fit)
res2<-resid(fit2)
plot(res2~yhat2)
abline(h=0,col=2,lty=2)
points(yhat2[Site == "a"],res2[Site == "a"],pch=20,col="red")
points(yhat2[Site == "b"],res2[Site == "b"],pch=20,col="blue")
legend("topright",legend=c("Site a", "Site b"),pch=20,col=c("red", "blue"))
```



```
fit3<-lm(logTEQ~Year+Site+Year*Site)
summary(fit3)
```

```
##
## Call:
## lm(formula = logTEQ ~ Year + Site + Year * Site)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.59906 -0.19601  0.00333  0.18401  0.51449
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  218.91364   44.64076   4.904 5.30e-05 ***
## Year         -0.10762    0.02236  -4.813 6.68e-05 ***
## Siteb        -95.67691   63.13157  -1.516  0.143
## Year:Siteb     0.04815    0.03162   1.523  0.141
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3373 on 24 degrees of freedom
## Multiple R-squared:  0.6434, Adjusted R-squared:  0.5989
## F-statistic: 14.44 on 3 and 24 DF,  p-value: 1.393e-05
```

Equivalent specification

```
lm(logTEQ~Year+Site+Year*Site)
```

```
##
## Call:
## lm(formula = logTEQ ~ Year + Site + Year * Site)
##
## Coefficients:
## (Intercept)      Year      Siteb  Year:Siteb
##   218.91364   -0.10762   -95.67691    0.04815
```

```
lm(logTEQ~Year*Site)
```

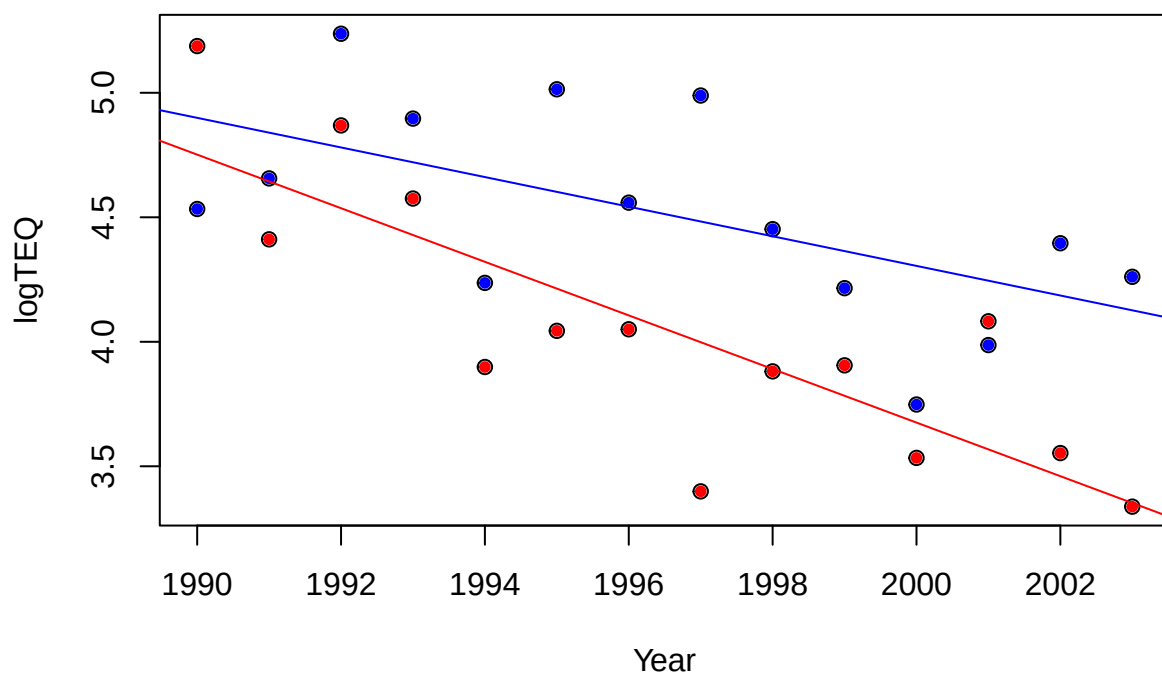
```
##
## Call:
## lm(formula = logTEQ ~ Year * Site)
##
## Coefficients:
## (Intercept)      Year      Siteb  Year:Siteb
##   218.91364   -0.10762   -95.67691    0.04815
```

```
lm(logTEQ~Year+Site+Year:Site)
```

```
##
## Call:
## lm(formula = logTEQ ~ Year + Site + Year:Site)
##
## Coefficients:
## (Intercept)      Year      Siteb  Year:Siteb
##   218.91364   -0.10762   -95.67691    0.04815
```

```
plot(logTEQ~Year,data=crabs)
points(Year[Site == "a"],logTEQ[Site == "a"],pch=20,col="red")
```

```
points(Year[Site == "b"], logTEQ[Site == "b"], pch=20, col="blue")
abline(coef(fit3)[1], coef(fit3)[2], col="red")
abline(coef(fit3)[1] + coef(fit3)[3], coef(fit3)[2] + coef(fit3)[4], col="blue")
```



```
anova(fit2, fit3)
```

```
## Analysis of Variance Table
##
## Model 1: logTEQ ~ Year + Site
## Model 2: logTEQ ~ Year + Site + Year * Site
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      25 2.9935
## 2      24 2.7297  1  0.26376 2.319 0.1409
```